RNA Extraction Using a Kit

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SUSTech-BIO 2026 iGEM Team

Purpose

Isolate total RNA from cultured cells or animal tissue using the FastPure Cell/Tissue Total RNA Isolation Kit V2. A gDNA-filter column precedes silica-column RNA purification; the eluate can be used for reverse transcription and subsequent gene-expression analysis. [1, 2]

Materials & Equipment

Item	Manufacturer	Identifier / specification	Use / Specification
FastPure Cell/Tissue Total RNA Isolation Kit V2	Vazyme	RC112-01	RL, RW1, RW2, RNase-free water, gDNA-Filter Columns III, RNA Columns III and collection tubes [1]
Ethanol	Sangon Biotech	Absolute grade	Also prepare 50% in RNase-free water for liver
RNase-free tips and tubes; pipettes	Axygen tips; laboratory equipment	RNase-free, v olume-compati ble tips	Sample handling and liquid transfer
Refrigerated microcentrifuge	Eppendorf	Compatible rotor; 13,400 × g	Use the protocol temperature
Vortex mixer	Kylin-Bell	Tube vortex mixing	Mix samples as directed
Homogenizer or mortar and liquid nitrogen	As available	RNase-clean tissue homoge nization	Tissue samples only
Temperature-controlled bath or block	As available	37°C; optional 65°C	Dissolving buffer crystals; optional warmed elution
DEPC-treated water	Sangon Biotech	B501005	External water for solution preparation; kit-supplied water for the kit workflow [3]

Before You Start

Use RNase-free consumables. Kit storage and routine extraction: 15-25°C. For the RC112-01 bottle containing 20 mL RW2 concentrate, add 80 mL absolute ethanol once, mix and mark the bottle. Dissolve any RL/RW1 crystals at 37°C; mix and return to room temperature. [1]

Sample	Input per extraction	Buffer RL	Ethanol after gDNA filtration
Cultured cells	<5 × 10^6 cells	500 µL	0.5 filtrate volume; absolute
Routine animal tissue	10-20 mg	500 µL	0.5 filtrate volume; absolute
Liver	≤10 mg	350 µL	1 filtrate volume; 50%
Kidney / spleen	≤10 mg	500 µL	0.5 filtrate volume; absolute

The input and reagent settings above follow the standard workflow. Highly fibrous or RNase-rich tissues require the manufacturer's sample-specific modifications; consult the linked manual before using those samples. [1, 2]

Procedure

Sample preparation

1. Remove culture medium. Lyse adherent cells directly, or pellet harvested/suspension cells and remove supernatant. Add RL according to the table and vortex until clumps disappear. For tissue, homogenize in RL on ice, or grind under liquid nitrogen before adding RL. [1]

Column purification

Follow the sequence below at 15-25°C. Every column spin uses 13,400 × g. [1]

2. Pass lysate through the gDNA-filter column for 30 s; retain filtrate and discard this column. Mix filtrate with the specified ethanol. Transfer the entire mixture, including any flocculent material, onto the RNA column. [1]

3. Complete RNA binding and washes in this order. Discard flow-through after each spin; retain the RNA column. [1]

Stage	Addition to RNA column	Spin at 13,400 × g
Binding	Entire ethanol/filtrate mixture	30 s
Wash 1	700 µL RW1	30 s
Wash 2	700 µL ethanol-prepared RW2	30 s
Wash 3	500 µL ethanol-prepared RW2	2 min
Conditional dry spin	None; empty collection tube first if residual liquid remains or the column touches flow-through	1 min

Elution

4. Move the RNA column into a clean RNase-free 1.5 mL tube. Apply 50-200 µL RNase-free water centrally without touching the membrane. Wait 1 min, then spin for 1 min. Retain eluate for downstream use or storage at -85 to -65°C. [1]

p.s.

Keep filtrate at the gDNA-removal stage, but keep the column during RNA binding and washing.

Choose centrifuge settings by relative centrifugal force. The manual's paired 12,000 rpm value is rotor-dependent.

For an optional yield improvement, prewarm elution water to 65°C and extend membrane incubation to 2-5 min before spinning. [1]

Expected Outcome

The final collection tube contains purified total RNA. Check concentration, purity and, when required, integrity before M10 reverse transcription. Record the input and elution volume alongside measurements; recovery varies with sample type and condition. [2]

RNA recovery alone does not establish absence of DNA. For expression assays, assess residual DNA with an appropriate no-reverse-transcriptase control.

Troubleshooting

Observed problem	Check first	Action
Slow flow / blockage	Input, clumps and tissue type	Reduce overloaded input; improve disruption. Use the manual's dedicated pretreatment for fibrous tissue. [1, 2]
Poor recovery	Sample condition, RW2 preparation and membrane wetting	Review input limits and ethanol preparation; use the optional warmed elution. [1]
Degraded RNA	Handling delays, thaw history and RNase exposure	Repeat with properly preserved material and RNase-free handling. [1]
Downstream inhibition	Wash sequence and liquid carryover	Check both RW2 washes and the conditional dry spin. [1]
Residual DNA	Input and use of the gDNA-filter column	Reduce excessive input in the next extraction. If further removal is needed, follow Vazyme's RH104-C1 DNase-set protocol. [2]

Reference

[1] Vazyme. FastPure Cell/Tissue Total RNA Isolation Kit V2, RC112, Instruction for Use, Version 24.2. Sections 02-06 and 08-09 (printed pp. 2-3, 5-8). Manufacturer-authored manual hosted by Biorbyt. Accessed 2026-09-19. Open source

[2] Vazyme Global. FastPure Cell/Tissue Total RNA Isolation Kit V2, RC112-01: product description, storage and FAQ. Accessed 2026-09-19. Open source

[3] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source